



Recycling utilization of Chinese medicine herbal residues resources: systematic evaluation on industrializable treatment modes

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Received: 27 May 2022 / Accepted: 24 January 2023 / Published online: 31 January 2023
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Abstract

Traditional Chinese medicine (TCM) is an indispensable part of the world health and medical system and plays an important role in treatment, prevention, and health care. These TCM produce a large amount of Chinese medicine herbal residues (CHMRs) during the application process, most of which are the residues after the decoction or extraction of botanical medicines. These CMHRs contain a large number of unused components, which can be used in medical, breeding, planting, materials, and other industries. Considering the practical application requirements, this paper mainly introduces the low-cost treatment methods of CHMRs, including the extraction of active ingredients, cultivation of edible fungi, and manufacture of feed. These methods not only have low upfront investment, but also have some income in the future. Furthermore, other methods are briefly introduced. In conclusion, this paper can provide a reference for people who need to deal with CMHRs and contribute to the sustainable development of TCM.

Keywords Chinese medicine herbal residues · Environment · Resource recycling · Industrializable · Pollution · Application

Introduction

The 2020 edition of the Chinese Pharmacopoeia records 2711 kinds of traditional Chinese medicine (TCM), a wide variety. In China, a great deal of CMHRs (Chinese medicine herbal residues) are produced in the process of using these TCM every year (Zhao and Tumenbayaer, 2019; Duan et al., 2019; Tao et al., 2021). In this paper, CMHRs refer to the solid biomass residue which is not the target product after the use of TCM. It can be the insoluble solid residue of TCM after decocting, or it can be the non-medicinal parts discarded during the use of traditional Chinese medicine (Saha and Basak, 2020). For example, the solid plant residues of herbal medicine after decocting and the non-medicinal

aboveground parts (stems and leaves) of TCM *Scutellaria*. Chinese patent medicine is a commercial TCM product made of a certain dosage form according to the prescribed prescription and preparation processes (Lu and Li, 2021). According to the National Bureau of Statistics, the national production of Chinese patent medicine in 2021 was about 2.5 million tons. The huge output represents a huge consumption of TCM, which further produces about 60–70 million tons of CMHRs each year (Huang et al., 2021). The main source of CMHRs is the solid plant residue discarded in the use of plant TCM, but there is also a small part of animal TCM and mineral TCM (Lu and Li, 2021).

Traditional CMHRs treatment methods include land-fill, incineration, and stack, which can cause high pollution (air, soil, and water) to the surrounding environment (Huang et al., 2021). It is well known that the incineration of CMHRs will produce carbon monoxide (CO) and carbon dioxide (CO₂), of which CO is an air pollutant, and CO₂ will aggravate the greenhouse effect and cause global warming. In addition, combustion will also produce acid gases such as nitrogen oxides (NO_x), HCl, SO₂, and HF, causing environmental problems such as acid rain, photochemical smog pollution, and urban haze weather (Lei et al., 2022; Ji et al., 2016). Moreover, CMHRs are perishable when stacked and landfilled, which in turn produce toxic odors to

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pollute the air and produce leachate to pollute water sources and groundwater (Qasim et al., 2020; Costa et al., 2019). Figure 1 is a sketch of environmental pollution caused by Chinese medicine herbal residues. Another important thing is that the above-mentioned treatment method will cause a huge waste of resources in addition to environmental pollution. Li et al. (2017) analyzed the nutrient content of six kinds of CMHRs of Ginseng nourishing plaster, Eguiyangxue granules, Duzhong Jiangya tablets, Astragalus granules, Gold Theragran, and Fuke Qianjin tablets. The results showed that the main nutrients contained in the CMHRs were crude protein, crude fiber, crude fat, amino acid, calcium, phosphorus, iron, magnesium, and so on. Moreover, many studies (Jasicka-Misiak et al., 2021; Wang, Liao, et al., 2020a; Wang, Wang, et al., 2020b) have proved that CMHRs contain unextracted active ingredients. (Both of them provide material basis for the reuse of CMHRs resources.) Therefore, only by using CMHRs rationally can we better protect the environment on which we live and make better use of resources.

It should be noted that the potential use of CMHRs depends on their chemical composition and physical characteristics. Because of the diversity of the TCM formula, the chemical composition of CMHRs is extremely complex (different chemical compositions exhibit different physical characteristics). Table 1 shows the contents of nutrients, amino acids, and major and trace elements of six kinds of CMHRs. It can be seen from Table 1 that the chemical composition of different drug residues differs greatly. For example, the

crude fiber content of RFQT is about 4 times higher than that of RGNP. The CMHRs with higher crude fiber content can be added to feed to promote the development of the digestive tract and stimulate gastrointestinal peristalsis (Li et al., 2017). The CMHRs with low ash content have the potential to prepare activated carbon, which can be further used in wastewater treatment (Saha et al., 2019). Some CMHRs containing active ingredients can be used for the direct extraction of active ingredients. For example, Saha et al. (2022) extracted flavonoids with antioxidant activity from the distilling waste of *Mentha arvensis*.

At present, the appropriate treatment methods of CMHRs include the extraction of active ingredients (Jasicka-Misiak et al., 2021; Wang, Liao, et al., 2020a; Wang, Wang, et al., 2020b), production of renewable energy (Ji et al., 2019; Yu et al., 2018), cultivation of edible fungi (Jin et al., 2018), manufacture of feed (Su et al., 2018), production of biological products (Qiu et al., 2020), and composting (Du et al., 2015). These methods have their own advantages and disadvantages in practical application and are suitable for different situations. Different from the previous review articles (Huang et al., 2021), this paper mainly discussed the extraction of active ingredients, cultivation of edible fungi, and manufacture of feed in consideration of the upfront costs and subsequent benefits. The aim is to provide a cheap, safe, and easy-to-implement CMHRs treatment scheme for small and micro pharmaceutical companies as much as possible. In addition, the production of renewable energy, composting, production of biological products, and other processing

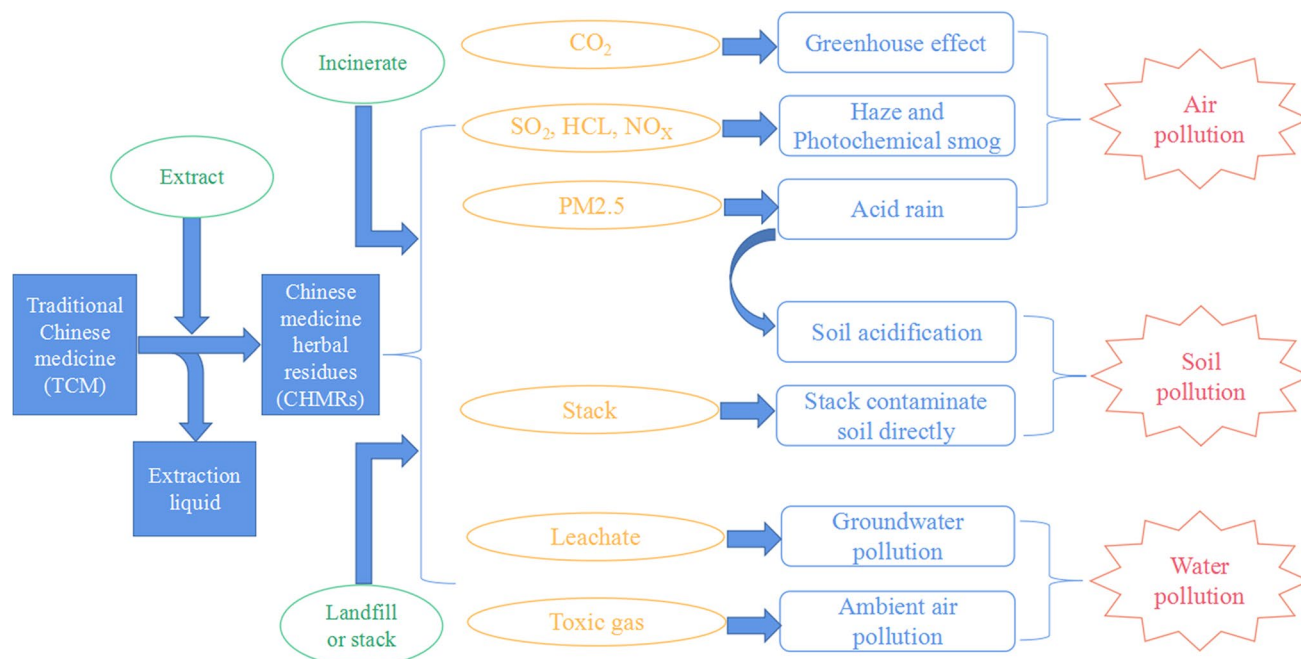


Fig. 1 The sketch of environmental pollution caused by Chinese medicine herbal residues

Table 1 Contents of nutrients, amino acids, and major and trace elements of six kinds of CMHRs (adopted from (Li et al., 2017))

Index	RGNP	REG	RDJT	RAG	RGT	RFQT
Dry matter (%)	93.48	92.65	96.07	93.41	93.24	90.44
Crude ash (%)	8.66	6.73	11.69	4.72	12.36	12.56
Crude protein (%)	10.50	12.10	10.74	10.97	16.46	11.97
Ether extract (%)	1.80	3.13	3.50	2.47	0.73	2.46
Crude fiber (%)	11.82	12.53	34.63	15.83	38.42	44.86
Acidic detergent fiber (%)	29.10	20.46	41.00	21.11	50.79	56.35
Neutral detergent fiber (%)	52.35	47.79	58.05	54.58	64.64	70.70
Phenylalanine (%)	0.68	0.37	0.30	0.28	0.40	0.52
Tyrosine (%)	0.40	0.25	0.20	0.19	0.26	0.32
Leucine (%)	1.26	0.64	0.56	0.48	0.69	1.04
Isoleucine (%)	0.80	0.26	0.27	0.22	0.33	0.70
Valine (%)	1.13	0.43	0.47	0.37	0.65	0.99
Alanine (%)	1.21	0.45	0.44	0.37	0.60	0.88
Glycine (%)	1.01	0.47	0.42	0.34	0.54	0.70
Aspartic (%)	1.80	0.78	0.73	0.72	0.96	1.37
Glutamic (%)	2.46	1.92	0.96	1.03	1.09	1.75
Proline (%)	0.83	0.45	0.40	0.37	0.48	0.77
Histidine (%)	0.37	0.18	0.17	0.15	0.21	0.42
Lysine (%)	0.51	0.38	0.30	0.32	0.48	0.82
Arginine (%)	0.58	0.33	0.42	0.49	0.33	0.47
Threonine (%)	0.89	0.39	0.36	0.31	0.47	0.71
Serine (%)	0.84	0.39	0.34	0.29	0.50	0.72
Methionine (%)	0.18	--	--	--	--	0.06
Total amino acids (%)	14.96	7.68	6.34	5.92	7.98	12.23
Ca (mg/kg)	8526.95	7683.26	9074.75	6625.62	15,995.50	23,880.68
P (mg/kg)	2075.32	2942.96	1572.68	2229.76	2340.22	1288.57
Fe (mg/kg)	1739.45	1387.11	1777.72	726.62	1729.47	2124.00
Mg (mg/kg)	2325.97	2979.01	3007.32	1843.76	5203.14	1748.76
Mn (mg/kg)	60.24	131.55	100.90	47.30	442.57	149.41
Pb (mg/kg)	6.48	--	2.19	1.60	4.33	3.30
Cd (mg/kg)	0.26	0.15	0.33	--	1.03	0.13

Note: Nutrient contents are based on the dry matter except of the dry matter

RGNP, residue of Ginseng nourishing plaster; REG, residue of Eguiyangxue granules; RDJT, residue of Duzhong Jiangya tablets; RAG, residue of Astragalus granules; RGT, residue of Gold Theragran; RFQT, residue of Fuke Qianjin tablets

methods are also introduced in this paper. Most of the articles cited in this review are articles in the past 5 years, which are relatively advanced ideas and technologies in the world and can well guide the rational reuse of CMHRs resources.

Extraction of active ingredients

Physical and chemical extraction

After the first extraction of TCM, there will be unextracted active ingredients in the remaining CMHRs, which can be extracted and reused by other different methods. Table 2 shows the application of CMHRs in the extraction of

active ingredients. Wang, Liao, et al. (2020a) extracted the *Periplaneta americana* residue polysaccharide (PAP) from the herbal residue of *P. americana* (75% ethanol extraction) with 10 times the amount of 0.02 M NaOH and then precipitated it with 3 times the amount of 95% ethanol to obtain crude PAP. The crude PAP was purified and freeze-dried to obtain PAP, and then carbomer (CBM) and carboxymethyl cellulose (CMC) were added to prepare a composite hydrogel for wound healing in diabetic rats. The results showed that the composite hydrogel could accelerate wound healing and relieve wound inflammation in diabetic rats. Li, Li, et al. (2019a) extracted a class of hemicellulose polysaccharide AX-I-3b from the herbal residue of *Astragalus radix* through a series of complex

Table 2 Application of CMHRs in the extraction of active ingredients

Feedstock	Active substance	Extraction method	Pharmacological action	References
Periplaneta americana herbal residue (purchased from Sichuan Good Doctor Panxi Pharmaceutical Co. Ltd.)	Periplaneta americana residue polysaccharide (PAP)	Alkali extraction and alcohol precipitation	Accelerates wound contraction, epidermis formation, collagen synthesis and deposition, angiogenesis, and reduces inflammatory response in diabetic rats	(Wang, Liao, et al., 2020a)
Astragalus radix herb residue (from the wild <i>Astragalus membranaceus</i>)	Hemicellulose Polysaccharide	Enzyme-Assisted Sodium Chlorite Extraction	Immunomodulatory activity and antitumor activity	(Li, Li, et al., 2019a)
Magnolia officinalis granules herb residue (from Beijing Kangrentang Pharmaceutical Co. Ltd.)	Magnolol and honokiol	60% ethanol reflux extraction	Inhibits colon cancer cells and protects against human neuroblastoma cell damage	(Zhou et al., 2021)
Polygonatum odoratum herb residue (from Chinese medicine chemistry laboratory, Shaanxi Institute of International Trade&Commerce)	Total flavonoids	Ultrasound-assisted alcohol extraction	Antioxidant capacity in vitro	(Wang, Wang, et al., 2020b)
Honeysuckle (from Anhui Bangtai Pharmaceutical Co. Ltd.)	Total flavonoids	Microblogging-assisted alcohol extraction	Anti-inflammatory and antioxidant	(He et al., 2020; Liu et al., 2020)
Industrial Panax notoginseng dregs (provided by Yunnan Yunke Pharmaceutical Co. Ltd.)	Dencichine	Water extraction and alcohol precipitation	Shorten clotting time	(Li, Yang et al., 2019b)
Panax ginseng residue (provided by Huarun Sanjiu Pharmaceutical Company)	Polysaccharides, ginsenosides, and succinic Acid	Enzyme-assisted n-butanol extraction and microbial fermentation (<i>Escherichia coli</i> strain ZW333)	Polysaccharides: Antioxidant, antitumor and immunomodulatory activities; ginsenosides: antidiabetic; succinic Acid: hypoglycemic and hypolipidemic effect	(Su et al., 2021; Zhou et al., 2019; Björklund et al., 1996)
Jianweixiaoshi tablets herb residue (from River Pharmaceutical Co. Ltd.)	Herb residue fermentation supernatant	Microbial fermentation	Treatment of <i>Helicobacter pylori</i> infection and diarrhea	(Meng et al., 2017a, 2017b)
Jianweixiaoshi tablets herb residue (from Jiangsu Yangtze River Pharmaceutical Group Company, Ltd.)	Herb residue fermentation supernatant	Microbial fermentation	Boost immunity	(Zhao et al., 2018)
Huazhenghuisheng oral-liquid residues (from Hayaocihang Pharmaceutical Co. Ltd.)	Nine medicinal anthraquinones	Microbial fermentation	Antibesity activity	(Kong et al., 2019)

extraction and purification. After experimental verification, the component has immunomodulatory activity and anti-tumor activity. Saha et al. (2022) obtained phenolic compounds and flavonoid compounds with antioxidant activity from the distillation waste biomass of *Mentha arvensis* by methanol extraction. Then, the optimum extraction solvent was 75% methanol based on extraction yield, total phenolic content, and flavonoid content. Similarly, Saha et al. (2021) also conducted a similar study on the distilled waste of Java citronella (*Cymbopogon winterianus* Jowitt) and reached a similar conclusion: phenols and flavonoids could be extracted from the distilled waste. In addition, Zhou et al. (2021) used the reflux method to extract magnolol and honokiol from *Magnolia officinalis* granule herbal residue; Wang, Wang, et al. (2020b) used ultrasonic-assisted extraction method to extract total flavones with antioxidant activity from the herbal residues of *Polygonatum odoratum*; He et al. (2020) used microwave-assisted extraction method to extract total flavones from the herbal residues of honeysuckle; Li, Yang, et al. (2019b) used water extraction and alcohol precipitation method to extract dencichine with hemostatic activity from *Panax notoginseng* herbal residue. All of the above studies can prove that extracting active ingredients from CMHRs is a feasible way to treat CMHRs.

Microbial fermentation extraction

The meaning of microbial fermentation extraction is the method of adding different strains to different CMHRs to obtain active metabolites. Meng et al. (2017a, 2017b) studied the therapeutic ability of the fermentation supernatant of the residue of Jianwei Xiaoshi Tablet (JXT) on *Helicobacter pylori* infection after adding different strains of bacteria for fermentation. The results showed that the fermented supernatant obtained from the herbal residues of the JXT after adding *Lactobacillus plantarum* and fermenting at 37 °C for 24–36 h had a high healing ability to *helicobacter pylori* infection and could reduce the inflammation response of gastric mucosa. Zhao et al. (2018) added *Lactobacillus plantarum*, *Bacillus subtilis*, and other probiotics into the residue of JXT for fermentation and obtained the fermentation supernatant, to explore its therapeutic ability on spleen-deficiency mice. The results prove that the fermented supernatant could increase the secretion of IL-2, IL-4, and IFN- γ in spleen-deficiency mice, that is, enhance the immune capacity of spleen-deficiency mice. Kong et al. (2019) added *Aspergillus cristatus* to Huazhenghuisheng oral-liquid herbal residue for fermentation and obtained nine anthraquinones with medicinal value, and the surplus fermentation substrate has the possibility of being used for composting (Fig. 2 shows the application idea).

Cultivation of edible fungi

As early as 2000 years ago, the ancient book “Lv's Spring and Autumn” in China had records about edible fungi. Up to now, edible fungi are still one of the favorite foods of Chinese people because of their cheap price and delicious taste. In addition, it is reported (Zhou, 2022) that China's annual output of edible fungi in 2020 will be more than 40 million tons, accounting for more than 70% of the world's annual output. Behind such a high yield is a relatively expensive cultivation cost. The cultivation substrate of edible fungi is mainly composed of cottonseed husk, sawdust, bran, and corncob (Zheng et al., 2019), and the average price is RMB 2000 per ton. Adding CMHRs to the cultivation substrate to reduce the dosage of other components can not only reduce the purchase cost of the cultivation substrate, but also increase the nutritional value and yield of the cultivated edible fungi (see Table 3.). Jin et al. (2018, 2020) studied the effects of adding CMHRs into the cultivation substrate of *Pleurotus ostreatus* with corncob as the main substance on the yield, nutrient content, and antioxidant activity of *P. ostreatus*. The results showed that adding CMHRs into the culture substrate could promote the growth of mycelia; shorten the culture period; improve the protein content, total phenol content, and yield of *P. ostreatus*; and improve the antioxidant activity. Özçelik and Pekşen (2007) found that adding a certain proportion of hazelnut husks into shiitake mushroom culture substrate could achieve higher yield than the control group (without adding hazelnut husks). Xia et al. (2021) studied the effect of adding CMHRs to *Ganoderma lucidum* cultivation substrate on the growth rate and yield of *Ganoderma lucidum*. The results showed that the addition of CMHRs could inhibit the growth of mycelium to a certain extent, but when the amount of CMHRs was 48%, the growth rate of mycelium was not significantly different from that of the control group, and the yield of fruiting bodies was increased. Similarly, Zhang et al. (2021) used CMHRs to replace the cottonseed husks in the cultivation substrate to cultivate *Tremella*. When the amount of CMHRs added was 30%, the results were similar to the above.

In addition, different CMHRs have different cultivation effects on edible fungi. Cai et al. (2020) added six kinds of CMHRs (*Astragalus*, *Chuanxiong*, *Atractylodes*, *Panax ginseng*, *Codonopsis*, and *Cistanche*) respectively to the *G. lucidum* cultivating substrate, and explored the effect of adding CMHRs on the growth of *G. lucidum*. The results showed that *G. lucidum* grown on the cultivation substrate with the addition of *Chuanxiong* medicinal residues had higher nutritional value and yield than the control group and other residue groups. This suggests that we should choose different CMHRs as matrix additives according

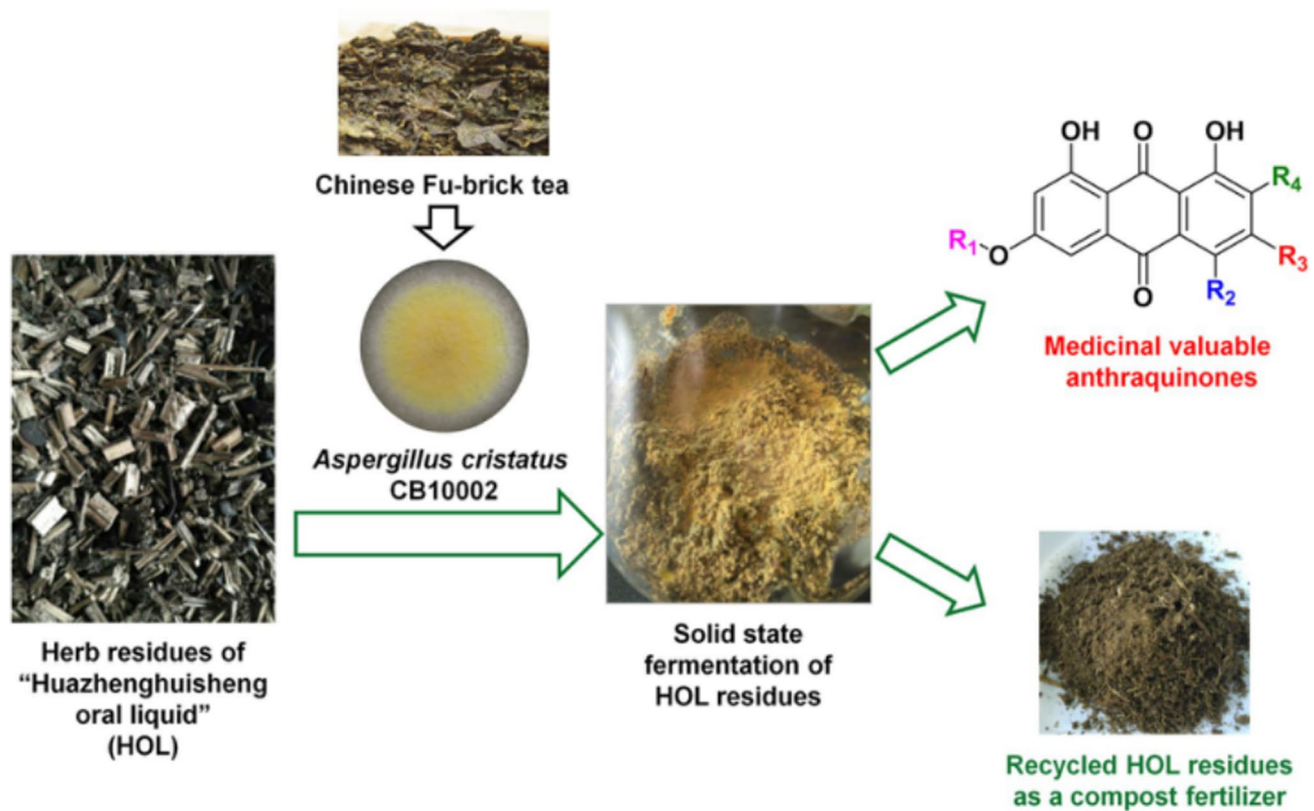


Fig. 2 The sketch for practical recycling of HOL residues by *A. cristatus* CB10002. (adopted from (Kong et al. 2019))

to the different nutrients required for the growth of edible fungi. The most important thing is that the waste left after edible fungi cultivation can be reused. The researchers (Zhou, Yan, et al., 2018b; Zheng et al., 2019; Guo, Song, et al., 2022a) believe that edible mushroom cultivation waste mainly has the following seven purposes, namely secondary cultivation, making soilless culture substrate, processing feed, making bio-organic fertilizer, producing energy substances, extracting active ingredients, and restoring the ecological environment.

Manufacture of feed

As early as centuries ago, herbal medicines have been widely used in the manufacture of feed, and their main role is to be added to feed as feed additives to play antibacterial, anthelmintic, antioxidant, growth promotion, and immune regulation roles (Kuralkar and Kuralkar, 2021). Kong et al. (2007) added TCM ultra-fine powder made of *Astragalus membranaceus*, *Acanthopanax senticosus*, *Codonopsis pilosula*, *Crataegus pinnatifida*, and *Salvia miltiorrhiza* into the feed of weaned piglets to explore the effect of TCM ultra-fine powder on weaned piglets. The results showed that TCM ultra-fine powder diet can increase the average daily feed

intake and weight gain of piglets, reduce the incidence of diarrhea, and enhance immunity. Liang et al. (2013) added 2% TCM (55% *A. membranaceus*, 27% *Angelica sinensis*, and 18% *Atractylodes lancea*) to sheep feed to explore the effect of TCM on sheep. The results showed that compared with the mixed hay group, the average daily weight gain of the sheep in the Chinese herbal medicine mixture group was significantly increased, and it had a positive effect on the rumen fermentation and energy metabolism of the sheep. However, with the increase in the price of TCM in recent years, the cost of adding TCM to feed is also increasing.

It is worth noting that CMHRs have great development potential in feed addition (Abdallah et al., 2019). Table 4 shows the application of CMHRs in the manufacture of feed. There are two main ways of application of CMHRs in feed: (1) The CMHRs are directly added to the feed; (2) the CMHRs are fermented and then added to the feed. Qu and Zhang (2015) studied the weight gain and economic benefits of feeding Hu sheep with CMHRs. The results showed that the fattening effect of the feed added with CMHRs was significantly higher than that of ordinary feed ($P < 0.01$), and the higher the content of CMHRs, the better the fattening effect. In addition, economic analysis shows that feeding Hu sheep with CMHRs feed can achieve higher economic benefits than feeding ordinary feed, and the average monthly net

Table 3 Application of CMHRs in the cultivation of edible fungi

Feedstock	Edible mushroom	Method of adding CMHRs	Effect	References
CMHRs				
Compound Kushen injection residues (CKI) and Qizhitongluo Capsule residues (from a pharmaceutical factory in Shanxi)	<i>Pleurotus ostreatus</i> (from the Mushroom Engineering Center of Shanxi Agricultural University)	CC + 30% CKI and CC + 30% QCI; CC + 30% QC2 and CC + 30% CKI.	Increase mycelial growth rate and increase yield	(Jin et al., 2018)
Compound Kushen Injection residues, Qizhitongluo Capsule (QTC) residues and Shenbai Shuxin Capsule (SSC) residues (from a pharmaceutical factory in Shanxi)	<i>Pleurotus ostreatus</i> (from the Mushroom Engineering Center of Shanxi Agricultural University)	CC + 45% CKI; CC + 60% QTC; CC + 30% SSC	Increase yield and nutritional value	(Jin et al., 2020)
Hazelnut husk (HH) (from hazelnut growers in the vicinity of Samsun province in Turkey)	<i>Lenitula edodes</i> (from Agromycel Company, Denizli, Turkey.)	25% HH + 75% wheat straw	Yield and bio-efficiency increased slightly	(Ozcelik and Pekşen, 2007)
Mixed CMHRs (from Shandong Zhongtai Pharmaceutical Co. Ltd.)	<i>Ganoderma lucidum</i> Karst (from Liaocheng Boyang Institute of Medicinal and Edible Fungi)	CMHRs 48%, mulberry wood chips 30%, bran 15%, corn flour 3%, gypsum 1%, quicklime 2%, superphosphate 1%	Increased spore powder and fruiting body production	(Xia et al., 2021)
Ganoderma lucidum Residue (from Fujian Zhongnan Pharmaceutical Co. Ltd.)	<i>Tremella fuciformis</i> (from Gutian County Hengchun Tremella Fungus Breeding Farm)	30% Ganoderma lucidum residue, 51% cottonseed husk, 18% bran, 1% gypsum	Increased yield and increased nutritional value	(Zhang et al., 2021)
<i>Astragalus</i> , Chuanxiong, <i>Atractylodes</i> , <i>Panax ginseng</i> , <i>Codonopsis</i> , and <i>Cistanche</i> (from Jiangxi Baishen Pharmaceutical Co. Ltd.)	G-I <i>Ganoderma lucidum</i> (self-made)	Cottonseed hulls 63%, CMHRs 20%, corn flour 10%, bran 5%, gypsum powder 1%, sucrose 1%	Increased yield and increased nutritional value	(Cai et al., 2020)

CC, corncoobs; QCI, consisted of Huangqi (Radix astragali) and Renshen (Radix Ophiopogonis), Wuweizi (Fructus Schisandrae chinensis), Danggui (Radix angelicae sinensis) and so on

Table 4 Application of CMHRs in manufacture of feed.

Feedstock	Processing method	Added amount	laboratory animal	Effect	References
<i>Astragalus membranaceus</i> , <i>Acanthopanax senticosus</i> , <i>Codonopsis pilosula</i> , <i>Crataegus pinnatifida</i> , and <i>Salvia miltiorrhiza</i> (from Changsha Pharmaceutical Co.)	Pulverized into ultrafine powder with an average particle size of 30µm	2 g/kg	Crossbred (Duroc × Landrace × Yorkshire) piglets, average age 21 days old, average weight 5.64 kg	Increase the average daily feed intake and weight gain of piglets and reduce the incidence of diarrhea and enhance immunity	(Kong et al., 2007)
<i>Astragalus membranaceus</i> , <i>Angelica sinensis</i> , and <i>Atractylodes lancea</i>	Mix in a 6:3:2 ratio	2% of mix hay feed	Crossbred (Corriedale × Suffolk) sheared sheep with an average age of 7 months and a weight of 29 ± 2 kg	Average daily gain was increased, and rumen fermentation and energy metabolism were positively affected in sheep.	(Liang et al., 2013)
Mixed CMHRs (from a Chinese medicine factory in Changzhou)	unmentioned	The first month average 1.8 kg/day, the second month average 2.5 kg/day	Crossbred (Hu sheep × Dorper sheep) sheep with an average age of 4 months	Meet the nutritional needs of Hu sheep and reduce the incidence of diseases in Hu sheep	(Qu and Zhang, 2015)
Radix Rehmanniae Preparata, Fructus Crataegi, Pericarpium Citri Reticulatae, Fructus Hordei Germinatus, and Radix Glycyrrhizae (from Hunan Sunaccord Biological Technical Co. Ltd. China.)	Mix in a 4:2:2:1:1 ratio	4 kg/t	Crossbred (Duroc × Landrace × large white) piglets, average age 21 days old, average weight 6.12 ± 0.05 kg	Improve the colonic ecosystem of piglets	(Su et al., 2018)
Mixed CMHRs (from A large Chinese medicine company in Jiangxi)	air-dried	88% CMHRs	Hu sheep ram, weight 35.0±2.0 kg	Increase the expression levels of immunoglobulins A, G, and M and increase the expression level of growth hormone	(Zhang et al., 2020)

income of each Hu sheep increases from 52.35 to about 120 RMB. Su et al. (2018) used the residues mixture of *Radix Rehmanniae Preparata*, *Fructus Crataegi*, *Pericarpium Citri Reticulatae*, *Fructus Hordei Germinatus*, and *Radix Glycyrrhizae* (respectively: 4:2:2:1:1) as the fermentation substrate for to obtain the fermentation of CMHRs. Then, the fermentation of CMHRs and the CMHRs were added into the weaned piglets' feed at the ratio of 4 kg/t to explore the effects of the two additives on weaned piglets. The results showed that adding fermented CMHRs or CMHRs into the feed could modify the colon ecosystem of weaned piglets. In addition, a recent article (Zhang et al., 2020) reported that feeding Hu sheep with fermented CMHRs can significantly increase the expression levels of immunoglobulins A, G, and M in Hu sheep and, at the same time, significantly increase the expression level of growth hormone. In a word, feeding fermented Chinese medicinal residues can enhance the immunity of Hu sheep and promote the growth of Hu sheep. It should be noted that not all CMHRs have these good effects. Eyng et al. (2015) added the extract residue of propolis to the feed of broiler chicks and studied the immune response. The results showed that the addition of 1–4% propolis extract to the diet of 1–21-day-old broilers did not improve immune parameters.

It should not be ignored that China's current "Feed Raw Materials Catalogue" and "Feed Additive Variety Catalogue" do not include the category of CMHRs. The exploration and research of using CMHRs to manufacture feed are mostly in the laboratory research stage and have not entered the industrial production and marketing stage. However, researchers are encouraged to conduct research related to the new feed and new feed additives. According to "the Regulations on the Administration of Feed and Feed Additives," before a new feed or feed additive is put into production, its developers or manufacturers should apply to the State Council for approval, and the National Feed Evaluation Committee should review its safety, effectiveness, and impact on the environment. After approval, new products are monitored for 5 years. Quantitative change causes qualitative change. Only abundant basic research can promote the emergence of new feed and feed additives based on CMHRs, to further alleviate the problem of environmental pollution and resource waste caused by CMHRs and promote the sustainable development of the environment and economy.

Other ways of reuse

Compost

Composting is a process of preparing organic fertilizer from various organic wastes. CMHRs contain a lot of protein, fiber, nitrogen, phosphorus elements, and other nutrients

and are excellent compost raw materials. Studies (Guo, Wu, et al., 2022b; Zhang et al., 2017) have shown that adding a certain proportion of CMHRs in the composting process of chickens and pig feces can significantly reduce antibiotic resistance genes (ARGs) in the composting process, thus reducing the spread of ARGs in the ecosystem. Both studies suggested that the change of microbial community structure was the main reason for the change of ARGs abundance. Zhou et al. (2016) mixed compost with CMHRs and food waste in different proportions and tested its antibacterial ability. The results showed that the addition of CMHRs could promote the maturity of compost, improve the resistance to the pathogenicity of compost, and form pathogenicity-resistant compost. The author (Zhou, Selvam, et al., 2018a) also composted food waste: sawdust:CMHRs with a ratio of 1:1:1 for 56 days, and then fertilized cress seeds with the available fertilizer. The results showed that the seed germination index was significantly increased to 157.2%. The above study fully demonstrated that adding a certain proportion of CMHRs in the composting process can improve the nutrient content of compost, accelerate the maturity of compost, and reduce potential human pathogens in compost.

Sewage treatment

For factories with sewage discharge requirements, sewage treatment has always been a headache. Fortunately, CMHRs show the advantage of being cheap and efficient in sewage treatment. Feng et al. (2020) investigated the removal ability of ammonium and phosphorus in swine wastewater by CMHRs and spent *Pleurotus ostreatus* substrate. The results showed that the adsorption capacity of ammonium and phosphate reached 1131.65 and 63.41 mg/g and 1631.79 and 62.58 mg/g, respectively (pH=8.0; dosage 0.2 g/L; adsorption time: 6 h). Among them, the removal principle of ammonium salt is mainly electrostatic attraction, ion exchange, and surface precipitation. The removal mechanisms of phosphate are ligand exchange, surface complexation, and surface precipitation. Zhao et al. (2016) studied the adsorption capacity of *Salvia miltiorrhiza* residue for methylene blue, a common harmful substance in textile industrial wastewater. And it turns out, *Salvia miltiorrhiza* residue has a certain adsorption capacity for methylene blue, and the citric acid and NaOH could enhance the adsorption capacity (from 100.00 to 161.29 mg/g and 178.67 mg/g). Saha et al. (2018) studied the residual waste plant materials after the distillation of *Palmarosa* (*Cymbopogon martinii*). The author (Saha) modified the previous method (Jain et al., 2013) and prepared alginate beads from the plant materials. Then, the adsorption capacity of the alginate beads to methylene blue in aqueous solution was studied. The results showed that the alginate beads could absorb methylene blue

in aqueous solution, and the maximum removal rate was 84% when the amount of adsorbent was increased from 1 to 10 g/L. Shang et al. (2017) produced a composite material using *Astragalus membranaceus* residue as raw material to remove chromium from acidic wastewater. The experimental results show that the composite material has a very high chromium removal rate (up to 98.71%). Saha et al. (2019) successfully prepared a new type of activated carbon (JCAC) from *Cymbopogon winterianus* solid residues after distillation through 85% phosphoric acid activation and oxygen-free pyrolysis in a muffle furnace. JCAC was used to remove the toxic anionic dye Congo red from wastewater and was compared with commercial activated carbon (CAC). The results show that JCAC has the ability to remove toxic anionic dye Congo red from wastewater, and compared with CAC, JCAC is only slightly inferior in terms of removal rate, but JCAC can still be used after 3 cycles and has higher repeatability than CAC. Correa-Navarro et al. (2020) prepared a biochar FB850-3 by pyrolysis of fique bagasse at 850 °C for 3 h. It was found that FB850-3 had the ability to absorb caffeine and diclofenac from water with the adsorption capacity of 40.2 mg/g and 5.20 mg/g, respectively. In addition, previous studies have proved that biochar prepared from the residue of the TCM *Evodia lepta* has the ability to absorb tetracycline in water (Cheng et al., 2019).

Production of renewable energy

The production of renewable energy mainly refers to the production of biogas and ethanol by fermentation, among which the production of biogas by fermentation is the method used by some large pharmaceutical companies to process CMHRs. It is reported (Tian, 2019) that China's Yiling Pharmaceutical used CMHRs for the preparation of biogas, which can process about 30,000 tons of CMHRs every year, producing about 4.6 million m³ of biogas. Created a very high economic value. Ji et al. (2019) explored the biogas production of CMHRs (obtained in the production of steroids) under different fermentation temperatures and different total solid contents. The results showed that when the temperature was 32 °C and the total solid content was 3.8%, the gas production from CMHRs was 667 mL/g, and in a certain range, the higher the temperature, the more gas production. Different CMHRs have different gas production performance. Xi et al. (2021) studied the effect of adding six different CMHRs (*Danshen* root, *A. membranaceus*, *Isatis* root, *A. sinensis*, pseudo-ginseng, and *Codonopsis pilosula*) on biogas production from fermentation. The results showed that the addition of CMHRs had a significant effect on anaerobic co-digestion, and the addition of 10% pseudo-ginseng had the greatest increase in biogas and methane production, increasing by 28% and 37%, respectively. In addition, the pretreatment of CMHRs can destroy

refractory cellulose and other components, thereby improving the gas production potential of medicinal residues. Yao et al. (2013) studied the changes of gas production capacity of CMHRs containing hawthorn, areca nut, *Aurantii trifoliata* after biogas slurry compost pretreatment, and NaOH pretreatment. The results showed that the gas production capacity of the CMHRs pretreated with 5% NaOH reached 196.8 L/kg after 10 days, which was significantly higher than that of 54.4 L/kg after compost pretreatment. In addition, fermentation to produce ethanol is another way to obtain renewable energy. At present, there are two common ethanol preparation processes, respectively: step saccharification fermentation process (Zhang et al., 2016) and synchronous saccharification fermentation process (Zhang et al., 2013). Liu (2020) has conducted research on the bio-refining of corn stalks to produce biofuel ethanol. First, the raw material is pretreated with acid sulfite to obtain the substrate, and then ethanol is produced by semi-synchronous saccharification and fermentation process. The results showed that when the substrate concentration was 15%, the ethanol yield reached 36 g/L after 48 h, the cellulose conversion rate was about 86%, and the ethanol production efficiency was about 0.752 g/L/h. In addition, previous studies have shown that *Akebia* herbal residues and post-harvest sugarcane residue can also be used to produce ethanol after treatment (Dawson and Boopathy, 2007; Yu et al., 2018).

Soil improvement

Biochar is a solid carbon-rich product obtained by pyrolysis and gasification of waste biomass at a high temperature (Yuan et al., 2019; Jiang et al., 2020). According to modern research, biochar is an excellent soil conditioner, which can improve soil condition and prevent pollution (Shaaban et al., 2018). CMHRs can be used as raw material for biochar production. Saha et al. (2022) prepared a biochar as a potential soil amendment from *Mentha arvensis* distillation residue and conducted a short-term soil culture study. The results showed that the biochar could effectively improve soil pH, cation exchange capacity, and N, P, and K content and improve soil microbial biomass carbon and dehydrogenase activity. Raw materials, pyrolysis temperature, and pyrolysis time are the key factors affecting the quality of biochar (Yuan et al., 2019). Saffari et al. (2020) studied the effects of biochar prepared by anaerobic pyrolysis of corn residue (stalks, cob, and straw) and poultry manure for 2 h at different temperatures on soil quality indicators and structural stability. The results indicated that when the pyrolysis temperature was 350 °C, the amount of corn residue biochar was 2%, and the amount of poultry manure biochar was 1%. The obtained biochar had the best soil improvement effect. In addition, biochar has a certain ability to repair many types of soil pollution. Xiao and Ding (2019) prepared biochar

from *Radix isatidis* residue by pyrolysis at 500 °C for 4 h. The study results show that the biochar has the ability to repair soil polluted by heavy metals Cu and Cd. Li and Li (2022) used straw as feedstock to prepare biochar for the treatment of saline–alkali land. It was found that straw biochar could significantly reduce soil salinity and promote soil phosphorus conversion. Recent studies have shown that biochar can also increase crop yields and reduce carbon dioxide emissions (Feng et al., 2021; Hu et al., 2019).

Production of industrial products

CMHRs are rich in cellulose, lignin, and other substances and are excellent raw materials that can be used to produce some biological products. Cellulose and lignin are excellent materials on their own. Wu et al. (2021) added gellan gum to the acid-hydrolyzed liquid of CMHRs for in situ fermentation to prepare bacterial cellulose. The results showed that adding gellan gum could increase the yield of bacterial cellulose from 0.543 to 0.866 g/L. Lignin can also be prepared from CMHRs. Using *Acanthopanax senticosus* residues as raw materials, Yu et al. (2021) investigated the effects of four extraction methods, namely, the alkali method, milled wood method, deep eutectic solvent (DES) method, and ethanol method, on the yield of lignin. As a result, the deep eutectic solvent method had the highest lignin yield, reaching 49.6%. In addition, CMHRs can also be used to produce high-value products such as cellulase and lipid. Qiu et al. (2020) studied the ability of fermentation of Chinese medicinal solid wastes such as *S. miltiorrhiza* and licorice to produce cellulase. The authors screened a strain of *Penicillium expansum* SZ13 and determined the best fermentation conditions: temperature 35 °C and determined the b rand ⁻¹, adding 5% of CMHRs, and 5% of seed solution. Under this condition, the peak period of enzyme production can be maintained for 5 days. Similar to cellulase, lipids are made by fermentation. Li et al. (2020) took the residue of Songling Xuemaikang capsule as the fermentation substrate and added *Rhodospiridium toruloides* for fermentation to produce lipids. The results showed that lipids could be produced and the mass fraction of lipids reached 33.6%.

Discussion

Economy, environment, and sustainable development

With sufficient technical support, the recycling of CMHRs can produce certain economic and environmental benefits and achieve sustainable development (Wang, 2014). Patel et al. (2018) analyzed the feasibility of pyrolysis of biochar from biosolids using the Aspen Plus process model and

performed an economic analysis. The results show that when the water content of biosolids is less than 50%, the pyrolysis system can generate additional heat for generating electricity to generate additional income. Moreover, from an economic point of view, the management cost of biosolids and the selling price of biochar are key factors. Mabrouk et al. (2018) also used Aspen Plus to simulate the process of producing catechol from lignin extracted from olive tree pruning. The results showed that when the plant had a capacity of 2544 kg of feedstock per day, the total investment was about \$4.9 million, and the catechol price was \$1100/t. In addition, an easily overlooked fact is that the recycling of CMHRs has reduced the emission of environmental pollutants and promoted the sustainable development of the ecological environment. Meng et al. (2010) studied the catalytic cracking technology of Chinese medicinal residue to produce fuel oil. The results of research show that when the catalyst is Al₂O₃, the pyrolysis temperature is 450 °C, and the catalyst amount is 10%, the fuel yield is up to 34.26%, and the fuel calorific value is up to 24.91 MJ/kg. And, every 1 t of CMHRs catalytic cracking can replace about 0.35 t of coal, reduce 1.29 t of carbon dioxide emissions, reduce 75.93% of waste CMHRs emissions, and produce a certain economic value (about 260 RMB).

The TCM industry refers to the sum of business units or individuals engaged in the production, service, or other economic behaviors related to TCM. Baidu Baike divides the TCM industry into Chinese herbal medicine and Chinese patent medicine. In essence, the recycling and utilization of CMHRs resources are to reuse the originally abandoned items, improve the utilization rate of resources, and create additional value for herb farmers and pharmaceutical enterprises. In addition, reasonable recycling and utilization of CMHRs can reduce environmental pollution, protect the ecological environment on which we live, and increase people's recognition of the TCM industry. Economic, environmental, and social are the three pillars of sustainable development (Zhu, 2019). Therefore, when the three aspects are satisfied, the recycling and utilization of CMHRs can certainly promote the sustainable development of the TCM industry.

Possible bottlenecks

It is not a simple thing to recycle CMHRs resources on a large scale, and there may be various bottlenecks and restrictions in its practical application. The first is the technical bottleneck. Depending on the complex classification system of TCM itself, CMHRs should also be classified so that different types of residues can play their roles separately (Wang et al., 2017). However, it is a pity that there are still few basic researches related to it, which cannot provide sufficient and reasonable theoretical and technical basis. In addition, there is a lack of practical case studies on the application of laboratory

technology to the problem of CMHRs recycling. Technology that has not been tested is immature technology. Researchers need to strengthen the relationship with pharmaceutical companies, promote the pilot work of CMHRs recycling, analyze the costs and benefits, and work out a CMHRs recycling technology that can not only produce good economic benefits but also reduce environmental pollution. The second is the social bottleneck. There are many similarities between CMHRs recycling and waste recycling. For most countries, a lack of sufficient practitioners, sufficient capital injection, and sound management system may be the problems encountered in the large-scale recycling of CMHRs (Saha and Basak, 2020; Kumar et al., 2017; Khan et al., 2022). In terms of practitioners, there is a lack of cheap labor and technical and management personnel with knowledge of CMHRs recycling. To solve this problem, it is necessary to improve the social recognition and salary of relevant employees and establish a complete talent training system. In terms of capital, sufficient capital injection is the driving force to promote the development of CMHRs recycling industry. Given the importance of CMHRs recycling, the industry may need more policy support from the state to attract investors' attention. Finally, it is very important to establish a perfect and efficient management system for the recycling and utilization of CMHRs. In my opinion, the management system may include the manufacture, classification, transportation, preservation, application, detection, sales, and after-sale parts of Chinese medicine residues. Of course, this is just my personal shallow point of view; its improvement needs the joint efforts of researchers all over the world.

Future research

There are many studies on the recycling of CMHRs, which also prove that CMHRs have a high application potential, but there are still some shortcomings. The following are some suggestions: (1) Strengthen the basic research on the recycling and utilization of CMHRs. Due to the diversity of TCM compounds, TCM is mostly used in combination, which leads to the complex composition of CMHRs which is not conducive to the large-scale recycling of CMHRs. This requires us to carry out more detailed basic research and even achieve "one kind of CMHRs corresponds to one kind of recycling method." (2) Try to find a new method of recycling and utilization of CMHRs. With the development of science and technology, we should not be limited to the existing recycling methods of CMHRs, we should try to explore new treatment methods. (3) To establish the recycling mode of CMHRs. It is very important to integrate the existing recycling methods of CMHRs, either internally or in combination with other industries, to establish a complete recycling mode of CMHRs. (4) Combine laboratory research with practical problems to solve real-world problems. The

experience of combining laboratory experiments with practice is very important. Researchers should pay attention to accumulating relevant experience, gradually popularize, and finally establish industry standards, to achieve the goal of industrial treatment of CMHRs.

Conclusions

For the leading companies in the TCM industry, they favor emerging treatment methods that can generate high-added value. But for small and micro enterprises, expensive equipment costs and method exploration costs are unacceptable. Therefore, this paper focuses on the introduction of low-cost methods for the treatment of CMHRs and briefly introduces other methods. It is boldly predicted that industrialized CMHRs treatment plant will be the mainstream treatment way in the future. It can make full use of CMHRs to avoid waste of resources and environmental pollution and promote the green cycle of the TCM industry chain.

Acknowledgements Thanks to all reviewers for their valuable comments on this article.

Materials availability The Materials section is not applicable.

Author contribution Jiahao Luo: first draft writing, material collection, and material analysis. Rui Yang: material collection and drawing. Feifei Ma: tabulation. Wenming Jiang: material analysis. Chunchao Han: manuscript revisions and comments. All authors contributed to the study's conception and design. All authors read and approved the final manuscript.

Funding This work was financially supported by Shandong Provincial Natural Science Foundation, China (No. ZR2020MH370), and Science and Technology Small and Medium-Sized Enterprises Innovation Ability Enhancement Project (2022TSGC2456).

Data availability The data is available in the manuscript.

Declarations

Ethical approval Not applicable

Consent to participate All authors participated in this work.

Consent for publication All authors allow the publication of this paper.

Competing interests The authors declare no competing interests.

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